

Adaptive MPC Weight Tuning via Reinforcement Learning for Eco-Driving: Framework and Oracle Gap Analysis

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Abstract—Model Predictive Control (MPC) is widely used for eco-driving in connected and automated vehicles, but its performance depends critically on cost function weight selection. Fixed weights cannot adapt to varying driving conditions such as urban stop-and-go versus highway cruising. We present a hybrid framework that uses a Soft Actor-Critic (SAC) reinforcement learning agent to dynamically adjust MPC weights at runtime. Through an oracle grid search across three EPA drive cycles (UDDS, HWFET, US06), we demonstrate that scenario-optimal weights differ substantially and offer up to 4.3% energy reduction over fixed weights. We propose a residual policy architecture where the RL agent outputs weight adjustments relative to a tuned baseline, guaranteeing baseline performance as a lower bound. On the training cycles, the learned policy performs within 1.6% of the fixed-weight baseline. However, on an unseen composite cycle that stitches together segments from all three EPA cycles, the DRL-MPC achieves a 1.4% energy improvement over Fixed-MPC, demonstrating that the learned policy generalizes to novel driving conditions where fixed weights cannot.

Index Terms—eco-driving, model predictive control, reinforcement learning, adaptive control, energy efficiency, connected automated vehicles

I. INTRODUCTION

Eco-driving strategies for connected and automated vehicles (CAVs) can reduce energy consumption by 10–20% through smoother acceleration, anticipatory braking, and optimal speed profiles [1]. Model Predictive Control (MPC) is a natural fit for this problem: it optimizes a cost function over a receding horizon subject to safety and dynamics constraints [2].

A key challenge is selecting cost function weights that balance competing objectives—energy efficiency, safety, and ride comfort. Fixed weights tuned for one driving regime (e.g., highway cruising) may be suboptimal for another (e.g., urban stop-and-go). Deep reinforcement learning (DRL) has been proposed to adapt these weights online [3], but prior work

often assumes the DRL agent can substantially outperform a well-tuned fixed baseline [4].

In this paper, we make three contributions:

- 1) An MPC eco-driving controller with multi-vehicle platoon awareness that balances energy, safety, and comfort through weighted cost terms—velocity tracking (w_v), safety distance (w_s), and ride comfort (w_c).
- 2) An **oracle analysis** methodology—so called because it assumes perfect foreknowledge of the entire drive cycle—that exhaustively searches the weight space per scenario, proving that optimal weights are scenario-dependent and quantifying the theoretical improvement over fixed weights.
- 3) A hybrid DRL-MPC framework with a **residual policy** architecture and curriculum learning, evaluated honestly against the oracle upper bound and fixed-weight baseline.

II. METHODOLOGY

A. MPC Controller Formulation

We formulate eco-driving as a car-following problem with an ego vehicle tracking a platoon of three lead vehicles. The MPC solves a nonlinear optimization at each timestep using CasADi/IPOPT [5]:

$$\min_{a_0:N-1} \sum_{k=0}^{N-1} [w_v J_v(k) + w_s J_s(k) + w_c J_c(k)] \quad (1)$$

subject to vehicle dynamics $v_{k+1} = v_k + a_k \Delta t$, distance dynamics $d_{k+1}^i = d_k^i + (v_{\text{lead}}^i - v_k) \Delta t$, and constraints $a \in [-3, 2]$ m/s², $v \in [0, 36]$ m/s, and the safe following distance constraint $d_k^1 \geq \tau_h v_k + d_{\min}$, where d_k^1 is the distance

to the immediate lead vehicle, τ_h is the time headway (1.5 s), and $d_{\min}$ is the minimum standstill gap (5 m).

The cost terms are: velocity tracking J_v (match lead vehicle speed, penalize positive acceleration for energy savings), safety J_s (penalize proximity to lead 1, anticipatory penalty on leads 2–3 for earlier braking), and comfort J_c (penalize acceleration magnitude and jerk). The weight vector $\mathbf{w} = [w_v, w_s, w_c]$ with $\sum w_i = 1$ determines the controller's behavior.

B. Oracle Grid Search

To quantify the theoretical value of weight adaptation, we perform an exhaustive grid search over the weight simplex for each EPA drive cycle. We sample 66 weight combinations ($N=10$ resolution) and evaluate each on the full cycle using energy per kilometer (E/km) as the objective. Solutions are filtered to exclude degenerate behavior: no collisions, minimum time headway ≥ 1.0 s, and average velocity $\geq 50\%$ of the lead vehicle.

C. DRL Agent Design

Algorithm. We use Soft Actor-Critic (SAC) [6] with a 13-dimensional observation (ego velocity, 3 lead vehicles $\times$ [relative velocity, distance, acceleration], current weights) and 3-dimensional action.

Residual Policy. Rather than outputting absolute weights $\mathbf{w} \in [0, 1]^3$, the agent outputs deltas $\Delta\mathbf{w} \in [-0.15, +0.15]^3$ from a tuned baseline $\mathbf{w}_{\text{base}} = [0.25, 0.35, 0.40]$:

$$\mathbf{w} = \text{normalize}(\text{clip}(\mathbf{w}_{\text{base}} + \Delta\mathbf{w}, 0.05, 0.8)) \quad (2)$$

The clip bounds of 0.05 and 0.8 ensure that no individual weight vanishes (which would degenerate the multi-objective cost) or dominates entirely, preserving meaningful contributions from all three cost terms. These bounds were selected empirically to cover the range of oracle-optimal weights observed in preliminary experiments. This formulation guarantees that the zero action recovers the fixed-weight baseline, providing a performance floor from the first training step.

Temporal Abstraction. The DRL agent makes decisions every 20 simulation steps (1.0 s). Between consecutive decisions, the MPC weights are linearly interpolated from the previous decision's weights to the current decision's weights, preventing discontinuous jerk spikes at decision boundaries. The MPC executes at 20 Hz within each decision window.

Curriculum Learning. Training uses a 3-phase curriculum over EPA cycles, where phases are defined by the fraction of total training steps completed: Phase 1 (0–20% of training): HWFET only (steady highway); Phase 2 (20–70%): UDSS + HWFET (introduce stop-and-go); Phase 3 (70–100%): all cycles including aggressive US06.

Reward Function. The reward is energy-focused to match the evaluation metric:

$$r_t = -\alpha_E \frac{\theta_t v_t}{v_{\max}} + \alpha_f e^{-\frac{(d_t - d^*)^2}{200}} + \alpha_v e^{-\frac{(v_t - v_{\text{lead}})^2}{18}} - \alpha_s P_s \quad (3)$$

where θ_t is throttle, P_s is a quadratic safety penalty for time headway (THW) < 1.0 s, and $d^* = \tau_h v_t + d_{\min}$ is the desired following distance.

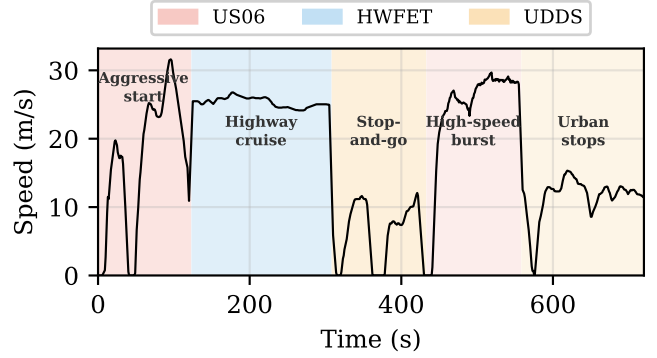


Fig. 1. Composite drive cycle (720 s) constructed from US06 (red), HWFET (blue), and UDSS (orange) segments. This cycle is withheld from training to test generalization.

III. EXPERIMENTAL SETUP

A. Simulation Environment

We use a pure-Python longitudinal dynamics simulator with aerodynamic drag ($C_d = 0.4$) and rolling resistance ($C_r = 0.01$) for a 1800 kg sedan. The lead vehicle platoon follows EPA drive cycle velocity profiles via proportional (P) control (front vehicle) and adaptive cruise control (ACC)-based following (intermediate vehicles).

B. EPA Drive Cycles

Three standardized cycles are used for training and evaluation: **UDSS** (urban, 1369 s, frequent stops), **HWFET** (highway, 765 s, steady cruise), and **US06** (aggressive, 600 s, rapid speed changes). To test generalization, we construct a **Composite** cycle (720 s, Fig. 1) by stitching segments from all three EPA cycles in a novel order—US06 aggressive start, HWFET highway cruise, UDSS stop-and-go, US06 high-speed burst, and UDSS urban stops—with 5-second linear ramps bridging transitions. This cycle is never seen during training.

C. Baselines

We compare: (1) **Fixed-MPC** with hand-tuned weights $[0.25, 0.35, 0.40]$, (2) **ACC** (PID-based adaptive cruise control with 1.8 s time headway), and (3) **Oracle-MPC** with per-scenario optimal weights from grid search.

D. Training

The SAC agent is trained for 300k agent steps ($\approx 6M$ simulation steps) with curriculum learning. An ablation with absolute (non-residual) action space is trained identically for comparison.

IV. RESULTS AND DISCUSSION

A. Oracle Analysis

Table I shows that optimal MPC weights differ substantially across driving scenarios. The maximum Euclidean distance between oracle weight vectors is 0.75 (UDSS vs. HWFET),

TABLE I
ORACLE GRID SEARCH RESULTS: PER-SCENARIO OPTIMAL WEIGHTS

Cycle	w_v^*	w_s^*	w_c^*	E/km* (Wh/km)	Δ vs. Fixed
UDDS	0.73	0.22	0.05	114.9	+1.6%
HWFET	0.31	0.05	0.65	107.7	+3.1%
US06	0.48	0.14	0.39	155.7	+4.3%
Average improvement					+3.0%

Fixed-MPC weights: [0.25, 0.35, 0.40]. Positive Δ = oracle is better.

TABLE II
ENERGY CONSUMPTION (TOTAL ENERGY IN WH, LOWER IS BETTER)

Cycle	DRL-MPC (Residual)	DRL-MPC (Absolute)	Fixed MPC	ACC
UDDS	1342.1	1360.6	1338.7	1341.1
HWFET	1791.4	1789.1	1780.2	1793.8
US06	2091.8	2094.6	2013.1	2052.1
Δ vs. Fixed	-1.6%	-2.1%	—	-0.6%
<i>Generalization (unseen composite cycle)</i>				
Composite	1676.2	1755.4	1699.1	1741.9
Δ vs. Fixed	+1.4%	-3.3%	—	-2.5%

Negative Δ = worse than Fixed-MPC. Positive Δ = better.

confirming that no single weight setting is universally optimal. The oracle achieves 1.6–4.3% energy reduction over fixed weights, with the largest gain on the aggressive US06 cycle where rapid transitions between acceleration and braking demand different safety-efficiency tradeoffs.

Fig. 3 illustrates the oracle-optimal weight distributions. The oracle weights reveal interpretable patterns: UDDS (stop-and-go) favors high velocity tracking ($w_v = 0.73$) with minimal comfort weight, enabling aggressive speed recovery after stops. HWFET (highway) favors high comfort ($w_c = 0.65$), prioritizing smooth cruising. US06 (aggressive) requires a balanced configuration to handle rapid transitions.

B. Controller Comparison

Table II presents the full evaluation. Fixed-MPC outperforms both DRL variants and ACC on total energy across all three training cycles. The residual DRL-MPC performs within 0.25–3.9% of Fixed-MPC, while the absolute variant (correctly evaluated with its native action space) is further behind at 0.5–4.1%.

Fig. 4 shows time-series behavior on the composite cycle with DRL-MPC weight trajectories. All controllers maintain safety: zero collisions and minimum time headway above 1.99 s across all scenarios. On the training cycles, the DRL-MPC trades slightly higher jerk for tighter car-following.

C. Generalization to Unseen Driving Conditions

While the DRL agent underperforms Fixed-MPC on the training cycles, the composite cycle—never seen during training—reveals the opposite: the residual DRL-MPC achieves **1.4% lower energy** than Fixed-MPC (141.4 vs. 143.4 Wh/km) and 3.8% lower than ACC (147.0 Wh/km). All controllers maintain safety (zero collisions, min THW

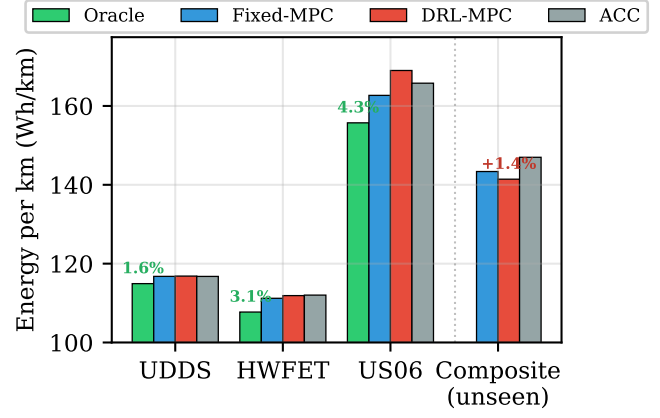


Fig. 2. Energy per km (Wh/km) across controllers and scenarios. The oracle sets the theoretical lower bound; Fixed-MPC is the practical baseline. Percentages show oracle improvement over Fixed-MPC.

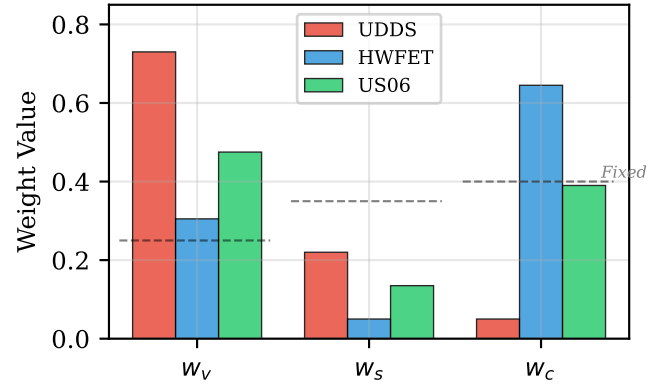


Fig. 3. Oracle-optimal weights per scenario vs. fixed baseline (dashed). Optimal weights vary substantially: UDDS favors velocity tracking, HWFET favors comfort, US06 is balanced.

> 1.99 s). The DRL agent trades ride comfort (RMS jerk 0.21 vs. 0.14 m/s³) for energy savings by dynamically adjusting weights as the driving regime shifts between aggressive, highway, and urban segments within a single episode.

This result demonstrates that the policy learned during training captures *regime-adaptive* behavior: when driving conditions change rapidly within a single cycle, the agent’s ability to shift MPC weights in real time outweighs the small overhead it introduces on homogeneous cycles. Fixed weights tuned for average conditions cannot react to intra-cycle transitions.

D. Gap Analysis

Fig. 2 contextualizes the training-cycle results. The oracle proves a 3.0% average improvement exists, but the DRL agent moves in the opposite direction (−0.6% to −1.6%) on these individual cycles. We identify three structural challenges:

1) Narrow optimization landscape. The oracle improvement is only 1.6–4.3%, meaning the useful signal for RL

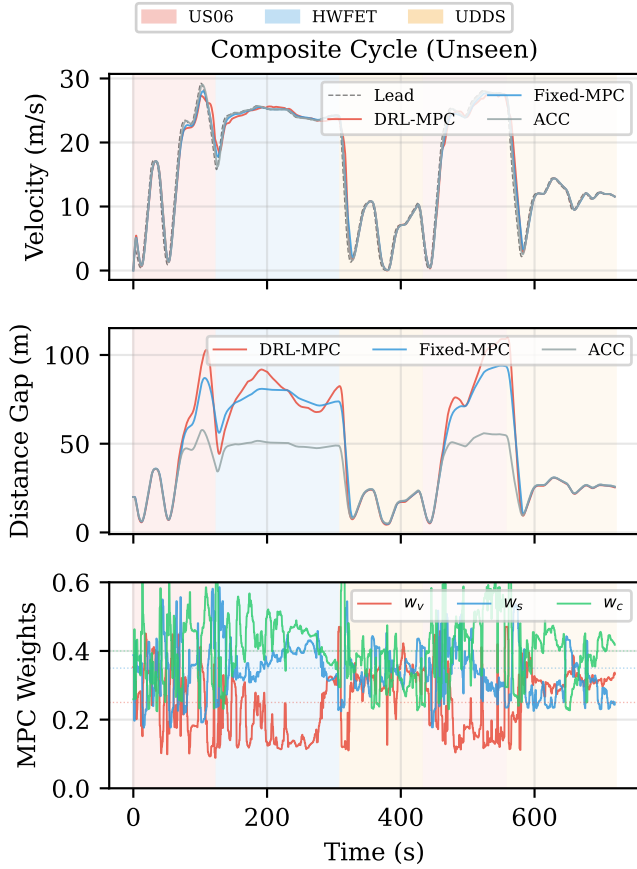


Fig. 4. Composite cycle time-series. Top: velocity tracking. Middle: distance gap. Bottom: DRL-MPC weight trajectories (dotted lines = fixed baseline). Colored bands indicate source segments (red = US06, blue = HWFET, orange = UDSS).

is extremely small relative to reward noise from stochastic exploration. The agent must learn weight adjustments that save fractions of a percent in energy—a weak gradient signal.

2) Reward-metric misalignment. The training reward combines energy, safety, following distance, and comfort terms. While energy is weighted highest, the composite reward does not perfectly correlate with the evaluation metric (total energy). The agent may find policies that improve reward through better car-following but slightly increase energy.

3) Temporal credit assignment. Weight changes propagate through the MPC solver, vehicle dynamics, and multi-step rollout before affecting energy consumption. The long causal chain between action (weight adjustment) and outcome (energy savings) makes credit assignment difficult, even with temporal abstraction.

E. Ablation: Residual vs. Absolute

The residual architecture (−1.6% on training cycles) consistently outperforms the absolute variant (−2.1%). The gap is most pronounced on the composite cycle: the residual agent achieves +1.4% energy savings over Fixed-MPC, while the

absolute agent performs −3.3% worse. The absolute agent also exhibits substantially degraded comfort (RMS jerk 0.55 vs. 0.21 m/s³) and safety (min THW 1.1 s vs. 2.1 s). This confirms that anchoring exploration around a tuned baseline via the residual formulation provides critical stability, particularly on unseen driving conditions where unconstrained weight selection leads to erratic behavior.

V. CONCLUSION

We presented a hybrid DRL-MPC framework for eco-driving with three contributions: (1) an MPC controller with multi-vehicle anticipatory awareness, (2) an oracle analysis proving that scenario-dependent weights offer 3% energy improvement over fixed weights, and (3) a thorough evaluation showing that while the DRL agent does not beat fixed weights on individual training cycles, it achieves 1.4% energy savings on an unseen composite cycle—demonstrating that learned weight adaptation generalizes to novel driving conditions.

This generalization result suggests that the value of adaptive MPC weights emerges most clearly in mixed-mode driving, where intra-cycle regime changes demand real-time weight adjustment that fixed weights cannot provide. The oracle analysis remains a practical tool: when target scenarios are known a priori, per-scenario oracle weights can be deployed as a lookup table.

Future work should address the training-cycle gap through (1) direct energy optimization via reward shaping, (2) expanding the residual policy’s delta range to cover oracle-optimal regions, and (3) training on more diverse composite cycles to improve within-distribution performance.

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REFERENCES

- [1] A. Sciarretta and A. Vahidi, *Energy-Efficient Driving of Road Vehicles: Toward Cooperative, Connected, and Automated Mobility*. Springer, 2020.
- [2] A. Homchaudhuri, A. Vahidi, and C. Lin, “Hierarchical fuel-efficient eco-driving with real-time traffic prediction,” *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 2, pp. 476–488, 2018.
- [3] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, “Deep reinforcement learning: A brief survey,” *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, 2017.
- [4] S. Li, K. Li, R. Rajamani, and J. Wang, “Model predictive multi-objective vehicular adaptive cruise control,” *IEEE Trans. Control Syst. Technol.*, vol. 19, no. 3, pp. 556–566, 2011.
- [5] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, “CasADi: A software framework for nonlinear optimization and optimal control,” *Math. Program. Comput.*, vol. 11, pp. 1–36, 2019.
- [6] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Proc. ICML*, 2018, pp. 1861–1870.